CSI HYDERABAD CHAPTER

DATA ANALYTICS USING POWER BI

Institutional Education Analytics System

Project Report

Submitted as part of CSI Internship

by

Sumanth Madham (MCA, 1st year)

Roll No: 25SA282031

GURU NANAK UNIVERSITY

Department of University institute of Computer Science and Applications(UICSA)

Under the guidance of

Dr.D.V.Ramana

Secretary, CSI

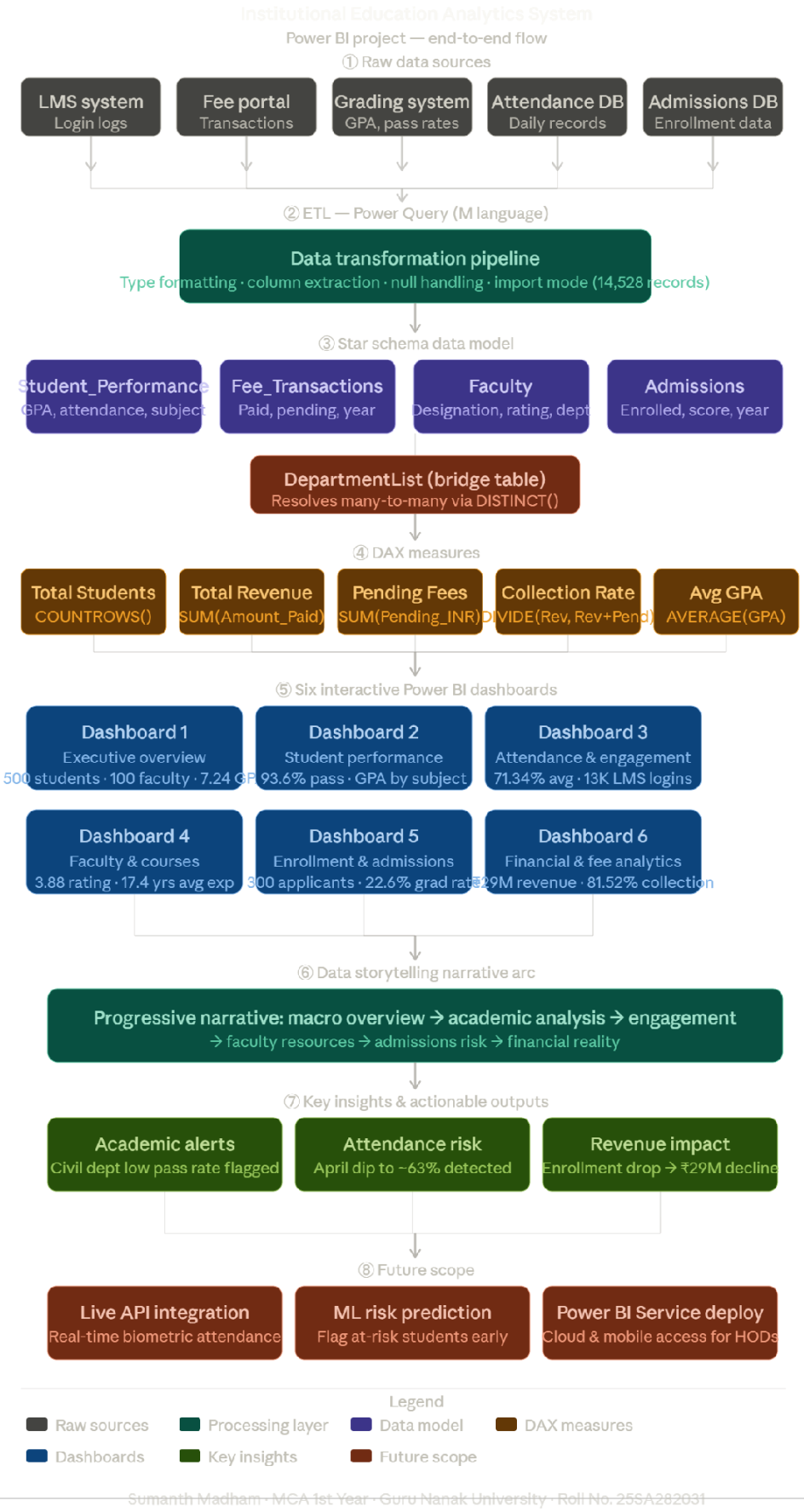
And

Dr.D.Krishna Kumar

Head of Department, UICSA

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Institutional Education Analytics System
Power BI project — end-to-end flow
① Raw data sources
LMS system
Login logs
Fee portal
Transactions
Grading system
GPA, pass rates
Attendance DB
Daily records
Admissions DB
Enrollment data
② ETL — Power Query (M language)
Data transformation pipeline
Type formatting · column extraction · null handling · import mode (14,528 records)
③ Star schema data model
Student_Performance
GPA, attendance, subject
Fee_Transactions
Paid, pending, year
Faculty
Designation, rating, dept
Admissions
Enrolled, score, year
DepartmentList (bridge table)
Resolves many-to-many via DISTINCT()
④ DAX measures
Total Students
COUNTROWS()
Total Revenue
SUM(Amount_Paid)
Pending Fees
SUM(Pending_INR)
Collection Rate
DIVIDE(Rev, Rev+Pend)
Avg GPA
AVERAGE(GPA)
⑤ Six interactive Power BI dashboards
Dashboard 1
Executive overview
500 students · 100 faculty · 7.24 GP
Dashboard 2
Student performance
93.6% pass · GPA by subject
Dashboard 3
Attendance & engagement
71.34% avg · 13K LMS logins
Dashboard 4
Faculty & courses
3.88 rating · 17.4 yrs avg exp
Dashboard 5
Enrollment & admissions
300 applicants · 22.6% grad rat
Dashboard 6
Financial & fee analytics
₹29M revenue · 81.52% collection
⑥ Data storytelling narrative arc
Progressive narrative: macro overview → academic analysis → engagement
→ faculty resources → admissions risk → financial reality
⑦ Key insights & actionable outputs
Academic alerts
Civil dept low pass rate flagged
Attendance risk
April dip to ~63% detected
Revenue impact
Enrollment drop → ₹29M decline
⑧ Future scope
Live API integration
Real-time biometric attendance
ML risk prediction
Flag at-risk students early
Power BI Service deploy
Cloud & mobile access for HODs
Legend
Raw sources
Processing layer
Data model
DAX measures
Dashboards
Key insights
Future scope

Introduction

In today's dynamic academic environment, institutional data is generated at an unprecedented scale across multiple departments. Administrators, faculty, and financial officers generate vast volumes of transactional data spanning student enrollment, attendance logs, academic performance, and fee collections. However, managing and deriving actionable intelligence from these fragmented datasets presents a significant operational and analytical challenge for university management.

This project delivers a comprehensive, interactive Institutional Analytics System Dashboard engineered using Microsoft Power BI. The dashboard consolidates and normalizes cross-departmental metrics covering hundreds of students, faculty members, and millions of rupees in financial transactions. By transforming raw academic and financial records into structured, interactive visual touchpoints, this business intelligence tool shifts decision-making from static spreadsheet reporting to dynamic, proactive institutional analysis.

The project builds a fully integrated enterprise-level analytics platform powered by Microsoft Power BI. It takes individual institutional metrics (GPA, attendance rates, pending fees, faculty experience) and delivers them through a common Star Schema data model with 6 interactive dashboard pages.

Scope

The scope covers all granular student, faculty, and financial attributes captured across the institution. The dashboard supports micro-analysis of variations across academic years (2022-2025), departments (MBA, Civil, Computer Science, etc.), and individual performance indicators. The integration of cross-filtering visual components provides data-driven administrative criteria.

Table of Contents

No.	Title	Page No.
1	Acknowledgements	2
2	Introduction	4
3	Problem Definition and Objectives	6
4	Power BI Overview	6
5	Story Telling Overview, Advantages & Challenges	8
6	Table Description – Columns & Data Types	9
7	Customized Columns and Measures	9
8	Dashboard 1 — Executive Overview	11
9	Dashboard 2 — Student Performance	12
10	Dashboard 3 — Attendance & Engagement	13
11	Dashboard 4 — Faculty & Course	14
12	Dashboard 5 — Enrollment & Admissions	15
13	Dashboard 6 — Financial & Fee Analytics	16
14	Application of Story Telling in the Project	17
15	Summary	17
16	Conclusion	18
17	References	19

Problem Definition and Objectives

Problem Overview

The Institutional Analytics System is an advanced Power BI solution designed to forecast academic trends and identify operational bottlenecks across the institution. It goes beyond traditional analysis by leveraging dynamic dashboards, scenario modeling, and real-time metric monitoring.

- **Data Fragmentation:** Institutional data is scattered across multiple sources (LMS, fee portals, grading systems) with inconsistent formats, making comprehensive analysis difficult.
- **Operational Opacity:** Lack of transparent, real-time academic and financial data leads to uninformed administrative decisions.
- **Performance Prediction Gap:** No reliable system exists to cross-reference student attendance directly with academic pass rates and fee collection efficiency.
- **Resource Risk:** Administrators cannot easily identify departments requiring immediate intervention without sophisticated tools.
-

Power BI Overview

Microsoft Power BI Desktop serves as the primary end-to-end business intelligence engine for this project's ETL pipelines, relational data modelling, Data Analysis Expressions (DAX) calculations, and interactive front-end layout.

What is Power BI?

Microsoft Power BI is a unified, scalable platform for self-service and enterprise business intelligence (BI). It connects to multiple data sources, transforms data, and creates interactive visualisations and reports to support informed decision-making.

- Power BI Desktop – Report authoring
- Power BI Service – Cloud publishing
- Power BI Mobile – On-the-go analytics
- Power BI Gateway – On-premises connectivity

Version Details

Component	Details
Power BI Desktop Version	2.139.x (June 2025 Release)
Power BI Service Deployment	Power BI Pro Capacity for internal organizational report sharing
DAX Engine Architecture	Analysis Services Tabular Model — xVelocity In-Memory Columnar Storage Engine for sub-second visual rendering
Power Query Engine	M Query Language 2.0 for data type formatting and column extraction
Native AI Integration	Key Influencers visual, Top Segments analysis, and AI-powered automated insights
Storage Mode	Import Mode — 14,528 records compressed in-memory for maximum DAX calculation speed

Key Power BI Capabilities Used

- **Import Storage Mode:** Leveraged to compress records into memory, maximizing calculation speed.
- **Cross-Visual Filtering:** Enabled across all 6 dashboard pages to preserve user analytical context.
- **Data Modeling (Star Schema):** Utilized a DepartmentList bridge table to resolve many-to-many relationship conflicts.
- **DAX Measures:** Custom measures for Average GPA, Total Revenue, Collection Rate, and Pass Rates.

Story Telling Overview, Advantages, and Challenges

Overview

Data storytelling is the analytical practice of building a structured narrative arc around data metrics to communicate complex property insights persuasively and effectively to investors and stakeholders. It effectively bridges three core pillars: **accurate data facts**, **clean interactive visualizations**, and **cohesive narrative context** that translates raw price numbers into direct investment decisions.

Advantages of Data Storytelling

 Clarity over Complexity: Translates multi-dimensional interactions into simple, scannable visual patterns.
 Accelerated Decision-Making: Highlights actionable insights immediately.
 Democratic Data Access: Equips non-technical administrators with self-service analytics capabilities through interactive slicers.

Challenges of Data Storytelling

- **Selection Bias Risk:** Curating visualizations selectively can lead to skewed narratives.
- **Data Quality Dependence:** A compelling storytelling flow remains vulnerable to poor upstream data engineering.

Table Description – Columns & Data Types

The foundational database schema consists of a consolidated data model. The table below outlines the structural schema:

Column Name	Data Type	Description
Academic_Year	Categorical (Text)	The academic cycle of the record (e.g., 2023-24).
Department	Text	The specific institutional department (e.g., MBA, Civil, Electronics).
Amount_Paid_INR	Whole Number	Processed fee amount in Indian Rupees.
Avg_GPA	Decimal Number	Academic performance metric.
Attendance_%	Decimal Number	The percentage of classes attended by the student.
Designation	Text	Faculty rank (e.g., Assistant Professor, HOD).

Customized Columns and Measures

Customized Columns and Measures

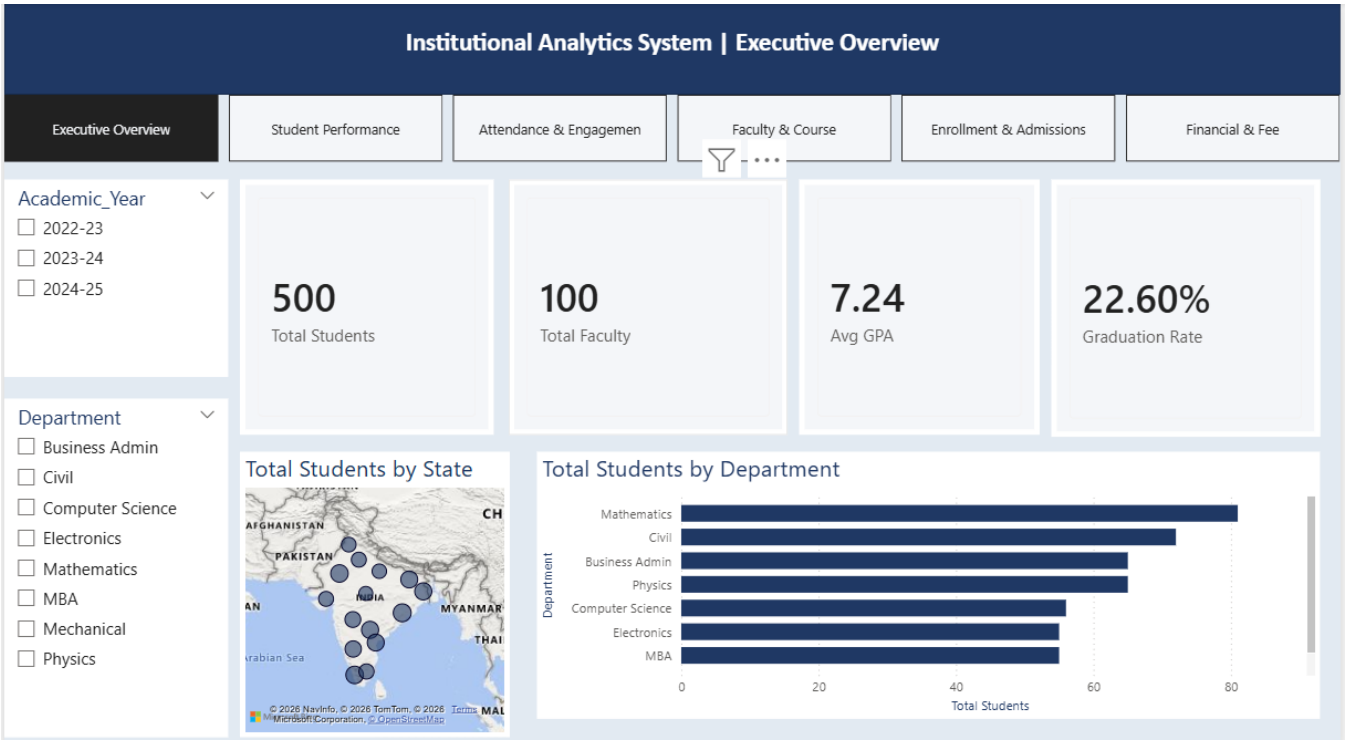
Calculated Columns Calculated columns were created to enrich the dataset with derived categorical attributes.

- DepartmentList: Created using DISTINCT(Faculty[Department]) to serve as a unified bridge table for cross-filtering.

Core DAX Measures Data Analysis Expressions (DAX) is a functional formula language that is the backbone of Microsoft Power BI.

Measure Name	DAX Formula
Total Students	COUNTROWS('Admissions')
Total Revenue	SUM('Fee_Transactions'[Amount_Paid_INR])
Pending Fees	SUM('Fee_Transactions'[Amount_Pending_INR])
Collection Rate %	DIVIDE([Total Revenue], [Total Revenue] + [Pending Fees], 0)
Avg GPA	AVERAGE('Student_Performance'[GPA])

Dashboard 1 — Executive Overview



Purpose This dashboard acts as a high-level executive summary of the entire institution. It aggregates core performance metrics to deliver an immediate health check.

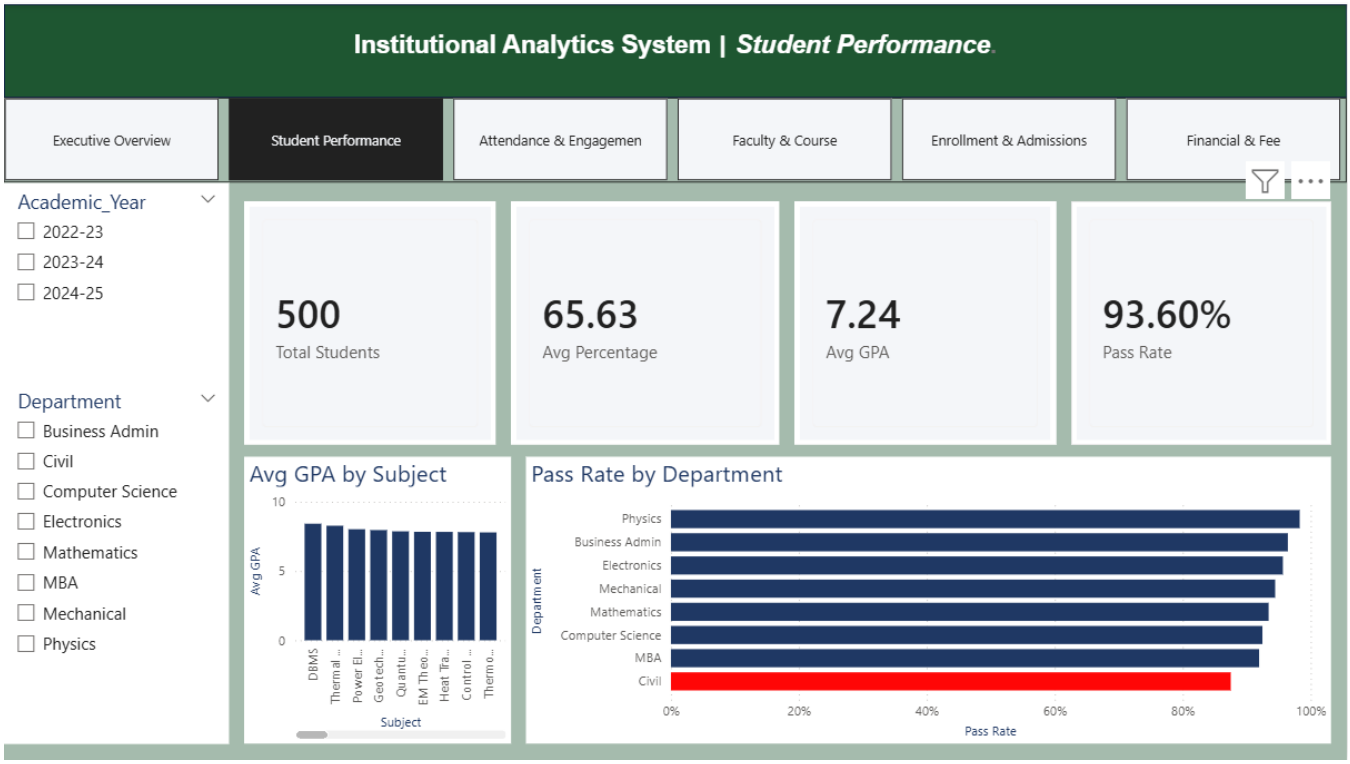
KPIs (Cards)

- **Total Students = 500:** Shows the overall enrollment.
- **Total Faculty = 100:** Indicates the teaching strength.
- **Avg GPA = 7.24:** Highlights overall academic standing.
- **Graduation Rate = 22.60%:** Shows the percentage of successful program completions.

Visuals & Scenario Analysis

- **Map Visual (Students by State):** Plots student origins geographically, helping admissions target specific regions.
- **Bar Chart (Students by Department):** Shows Mathematics and Civil leading in student count, which helps in resource allocation.

Dashboard 2 — Student Performance



Purpose Analyzes academic performance trends across departments and subjects.

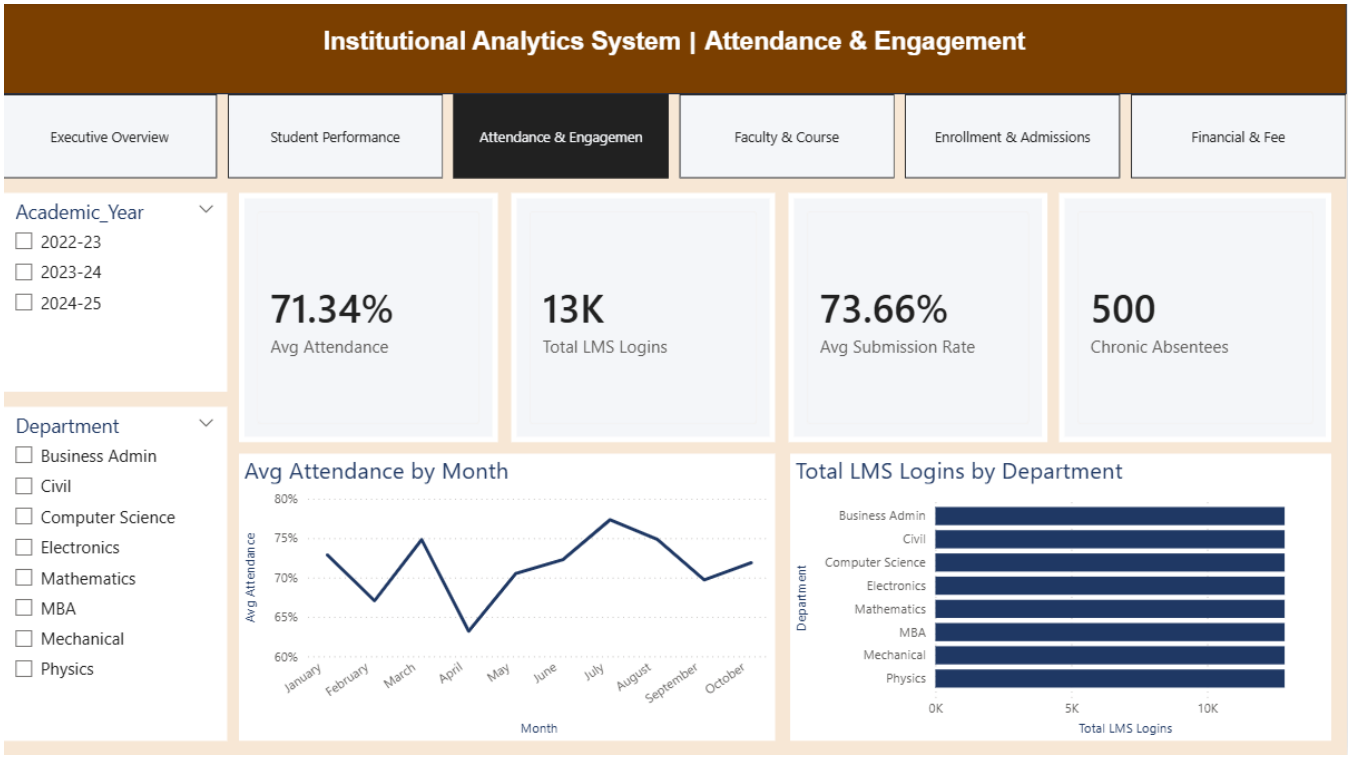
KPIs (Cards)

- **Total Students = 500**
- **Avg Percentage = 65.63**
- **Avg GPA = 7.24**
- **Pass Rate = 93.60%:** Indicates high overall institutional success.

Visuals & Scenario Analysis

- **Column Chart (Avg GPA by Subject):** DBMS and Thermal Engineering show strong GPAs.
- **Bar Chart (Pass Rate by Department):** Physics leads at near 100%, while Civil is highlighted in red, indicating a lower pass rate that requires immediate academic intervention.

Dashboard 3 — Attendance & Engagement



Purpose Deep-dives into student participation, measuring physical attendance and digital LMS engagement.

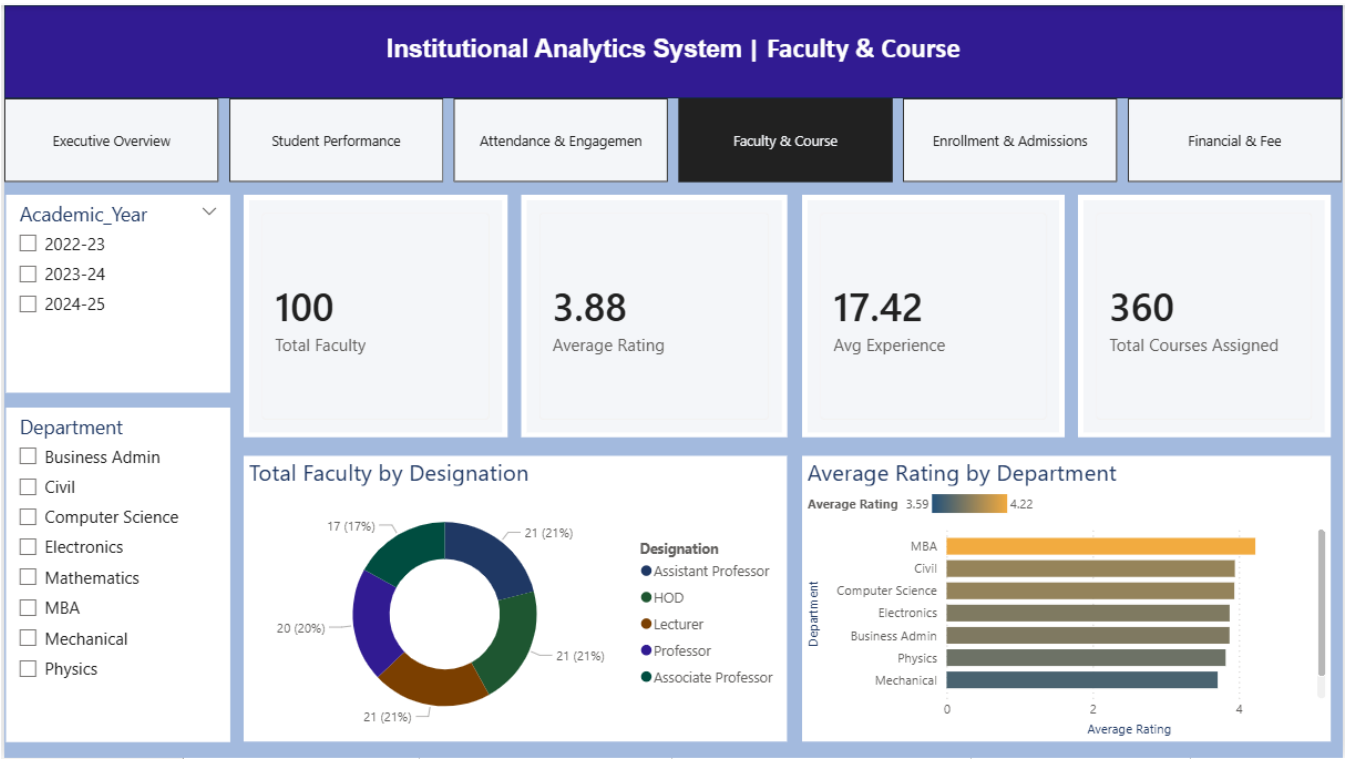
KPIs (Cards)

- **Avg Attendance = 71.34%**
- **Total LMS Logins = 13K:** Measures digital engagement.
- **Avg Submission Rate = 73.66%**
- **Chronic Absentees = 500**

Visuals & Scenario Analysis

- **Line Chart (Avg Attendance by Month):** Shows a severe dip in April (down to ~63%), prompting administrators to investigate causes (e.g., holidays, exam prep).
- **Bar Chart (LMS Logins by Dept):** Shows a relatively uniform distribution of digital engagement across all departments (~13K each).
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Dashboard 4 — Faculty & Course



Purpose Analyzes teaching resources, experience, and student feedback.

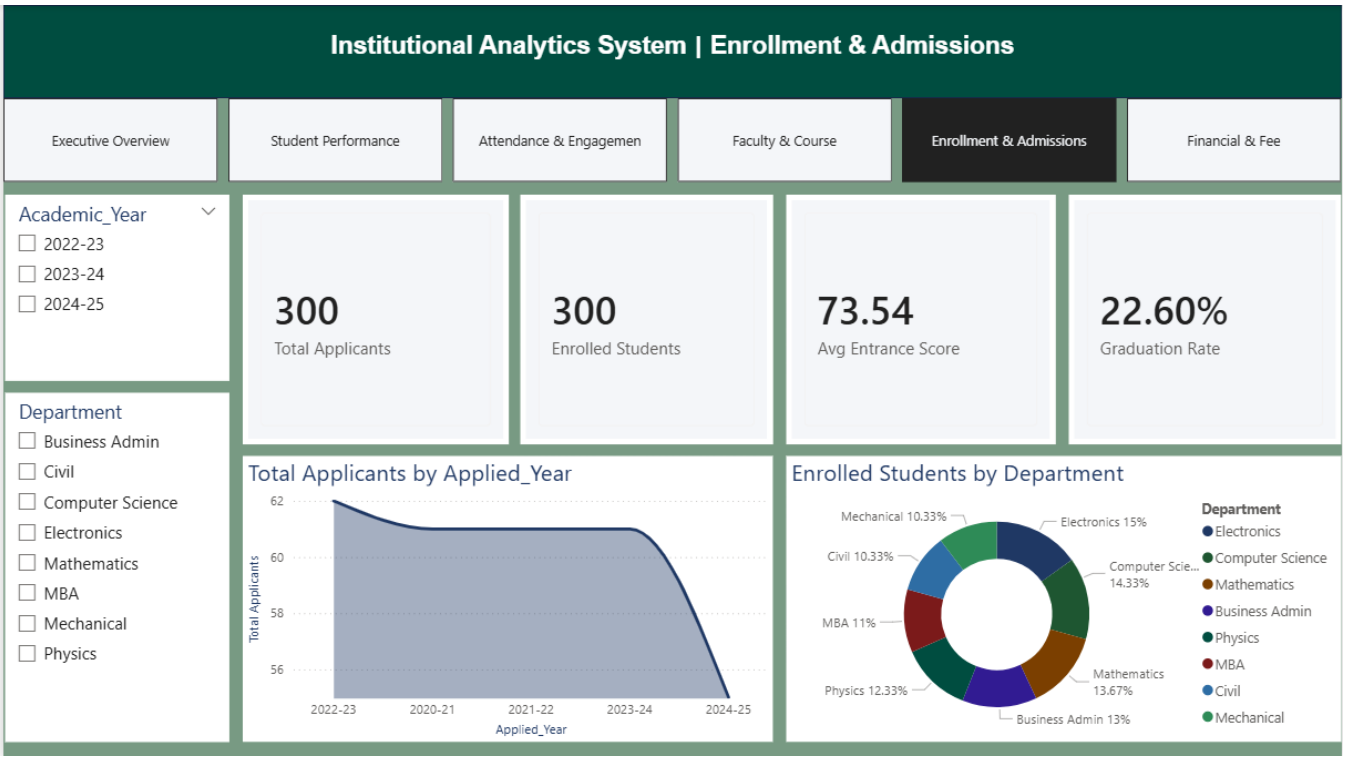
KPIs (Cards)

- **Total Faculty = 100**
- **Average Rating = 3.88:** Baseline for teaching quality.
- **Avg Experience = 17.42 Years**
- **Total Courses Assigned = 360**

Visuals & Scenario Analysis

- **Donut Chart (Faculty by Designation):** Shows a highly balanced hierarchical structure (21% HOD, 21% Assistant Prof, 20% Lecturer).
- **Bar Chart (Avg Rating by Dept):** MBA leads with the highest student ratings, serving as a benchmark for other departments like Mechanical.

Dashboard 5 — Enrollment & Admissions



Purpose Tracks the admissions funnel and departmental popularity.

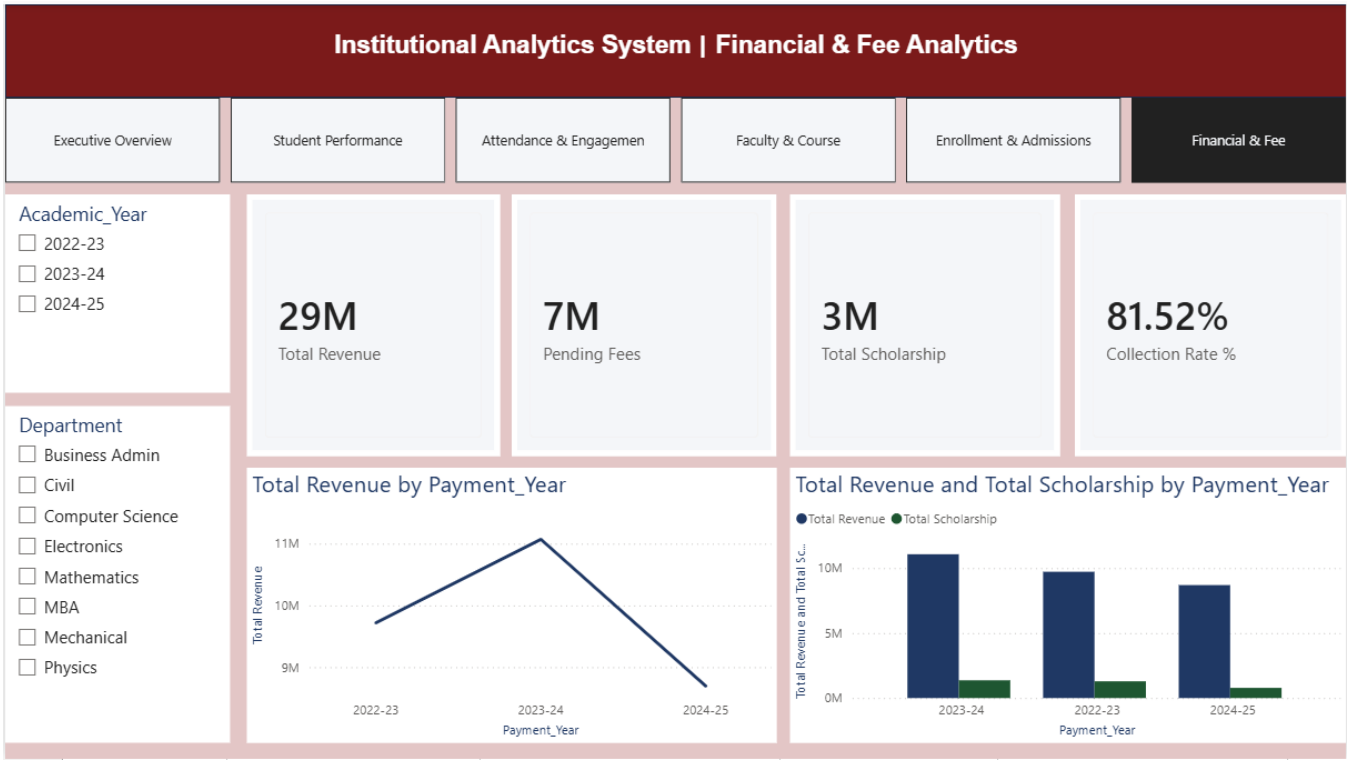
KPIs (Cards)

- **Total Applicants = 300**
- **Enrolled Students = 300:** Shows a 100% conversion rate for processed applications.
- **Avg Entrance Score = 73.54**
- **Graduation Rate = 22.60%**

Visuals & Scenario Analysis

- **Area Chart (Applicants by Year):** Shows a steady application rate that drastically drops heading into 2024-25, highlighting an urgent need for marketing intervention.
- **Donut Chart (Enrolled Students by Dept):** Electronics (15%) and Computer Science (14.33%) occupy the largest market shares.

Dashboard 6 — Financial & Fee Analytics



Purpose Provides budget-planning and financial benchmarking capabilities by tracking institutional revenue and deficits.

KPIs (Cards)

- **Total Revenue = 29M:** Overall capital generated.
- **Pending Fees = 7M:** Highlights outstanding debt.
- **Total Scholarship = 3M:** Institutional financial aid provided.
- **Collection Rate = 81.52%:** Efficiency metric for the finance department.

Visuals & Scenario Analysis

- **Line Chart (Revenue by Payment_Year):** Shows revenue peaked in 2023-24 (above 11M) but has sharply declined in 2024-25, mirroring the admissions drop.

- **Clustered Column Chart:** Compares Revenue vs. Scholarship side-by-side, proving that scholarships remain a stable proportion of revenue year-over-year.

Application of Story Telling in the Project

The dashboard pages are intentionally sequenced as a progressive narrative arc — moving the administrator from macro-level overviews down to specific academic and financial insights.

Phase	Dashboard	Narrative Role
Phase 1	Executive Overview	Sets the Scene — KPI cards establish the scale of the student body and faculty.
Phase 2	Student Performance	Deepens Narrative — Identifies which departments are succeeding academically and which are failing.
Phase 3	Attendance	Adds Context — Correlates engagement levels (April drop) with performance outcomes.
Phase 4	Faculty	Evaluates Resources — Benchmarks teaching quality across departments.
Phase 5	Enrollment	Highlights Risk — Visualizes the sharp decline in incoming applicants.
Phase 6	Finance	Concludes the Impact — Directly ties the drop in enrollment to the severe dip in institutional revenue.

Summary

The Institutional Analytics System project successfully demonstrates the power of Microsoft Power BI in transforming raw academic records into a coherent, actionable intelligence platform. Key project deliverables include:

- **Dashboard Pages:** 6 interactive Power BI pages.
- **DAX Measures:** Custom measures for GPA, Revenue, Pending Fees, and Collection Rates.
- **Data Modeling:** Successfully resolving many-to-many relationship conflicts using bridge tables.
- **Story Telling:** Narrative flows from academic overview → student performance → engagement → faculty resources → admissions tracking → financial reality.

Conclusion

The project successfully demonstrates the effectiveness of Microsoft Power BI in converting raw and unstructured educational data into meaningful insights. Various data preprocessing techniques were performed to improve data quality. Advanced DAX measures were developed to calculate important business metrics such as Total Revenue and Collection Rates. The interactive dashboards, utilizing slicers, filters, and cross-visual interactions, allow users to explore data dynamically. Overall, the project

demonstrates how Business Intelligence and Data Analytics can be effectively applied in the educational domain to support strategic decision-making and operational trend prediction.

Future Scope:

- Integrate live API feeds for real-time attendance tracking via biometric scanners.
- Add predictive ML models to flag "at-risk" students likely to fail or drop out.
- Deploy to Power BI Service for cloud-based, mobile-accessible sharing with HODs.

References

[1] Microsoft Power BI Documentation — https://learn.microsoft.com/en-us/power-bi/
[2] DAX Reference Guide – Microsoft — https://learn.microsoft.com/en-us/dax/
[3] Data Storytelling with Power BI – Alberto Ferrari & Marco Russo — https://www.sqlbi.com/
[4] Power BI DAX Patterns – SQLBI — https://www.daxpatterns.com/

Appendix – A: Customized Columns Created Using DAX

1. Department Bridge Table

- **Purpose:** Resolve relationship duplication across fact tables.
- **DAX Formula:** DepartmentList = DISTINCT(Faculty[Department])

Appendix – B: DAX Measures Used in the Project

1. Total Revenue Total Revenue = SUM(Fee_Transactions[Amount_Paid_INR])
Purpose: Calculates overall collected capital.
2. Pending Fees Pending Fees = SUM(Fee_Transactions[Amount_Pending_INR])
Purpose: Tracks outstanding student debt.
3. Collection Rate % Collection Rate = DIVIDE([Total Revenue], [Total Revenue] + [Pending Fees], 0) Purpose: Calculates percentage of total target fees successfully collected.
4. Average GPA Avg GPA = AVERAGE(Student_Performance[GPA]) Purpose: Calculates average institutional academic performance.

Appendix – C: Charts Used in the Project

1. Card Visual

- **Purpose:** Display key metrics such as Average GPA, Total Revenue.
- **Advantages:** Easy to understand, highlights important values.
- **Disadvantages:** Shows only one value at a time.

2. Donut Chart

- **Purpose:** Show distribution of Faculty Designations.
- **Advantages:** Good for percentage comparison.

3. Bar & Column Charts

- **Purpose:** Compare pass rates and GPAs across departments.
- **Advantages:** Easy category comparison.

4. Line Chart

- **Purpose:** Show average attendance and revenue trends over time.
- **Advantages:** Easy to identify trends.

5. Map Visual

- **Purpose:** Display geographical origin of students.
- **Advantages:** Provides location-based insights.

6. Slicer

- **Purpose:** Filter data by Academic Year and Department.
- **Advantages:** Interactive and user-friendly.